

# Arthur Marmin

## Curriculum Vitae

### RESEARCH TOPICS

- Convex and non-convex optimization
- Matrix and tensor decomposition
- Polynomial/Rational optimization
- Inverse problems for signal processing and PDE
- Sparse signal reconstruction

### WORK EXPERIENCE

SEPTEMBER 2022 — NOW  
Université Aix-Marseille, I2M  
**Assistant Professor**

Department of Mathematics, ALEA team

FEBRUARY 2021 — AUGUST 2022  
Signals and Communication Team, IRIT, Toulouse  
**CNRS Post-doctorate Researcher**

Study of optimization problems related to tensor decompositions

SEPTEMBER 2017 — JUNE 2020  
CentraleSupélec  
**Teaching Assistant**  
Optimization labs (2nd year) and reinforcement classes (1st year)

APRIL 2015 — APRIL 2017  
Industrial Technology Research Institute, Taiwan  
**Research Engineer**

Department System and Architecture Design

- Model power consumption and performance of embedded GPU, and write OpenGL and Vulkan benchmarks.
- Develop a collaborative software platform for SoC, EDA and IP designers to perform early power and performance analysis and validation.  
(Collaboration with TSMC)
- Implement image rendering technique on FPGA to reduce the motion-to-photon delay inside virtual reality devices.  
(Collaboration with University of California Santa Barbara)

FALL SEMESTER 2014  
National Chiao Tung University, Taiwan  
**Teaching Assistant (Embedded System Design)**

JULY — SEPTEMBER 2012  
TeamWork Vietnam, Việt Nam,  
**SAP Administrator**

🏠 Aix-Marseille Université, Campus Marseille Centre  
3 Place Victor Hugo, 13003 MARSEILLE  
Bureau 8-1/O-06  
☎ (+33) 7 69 96 86 40  
✉ arthur.marmin@univ-amu.fr  
💻 <https://www.arthurmarmin.github.io>

### FORMATION

DÉC. 2020 **Doctor of Philosophy in Signal Processing**  
Centre de Vision Numérique  
Advisors: Jean-Christophe Pesquet, Marc Castella  
Thesis: Rational models optimized exactly  
for solving signal processing problems  
OCT. 2017 *Université Paris-Saclay, CentraleSupélec*

JAN. 2015 **Master of Science**  
Electrical Engineering and Computer Science  
Advisor: Shiao-Li Tsao  
Thesis: Design and Implementation of an  
Embedded GPU Simulator

SEP. 2013 *National Chiao Tung University, Taiwan*

JAN. 2015 **Engineer Degree in Telecommunication**  
SEP. 2011 *Télécom SudParis*

### COMPUTER SKILLS

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| PROGRAMMING     | C++, Assembly Languages, Python,                                                     |
| HARDWARE DESIGN | Verilog, SystemC, TLM                                                                |
| API             | OpenGL, OpenCL, Qt, mpack,<br>Numpy, Scikit-learn, Keras<br>Linux kernel             |
| LOGICIEL        | Matlab, GloptiPoly, Tensorlab,<br>Xilinx Vivado Design suite,<br>Git, Doxygen, Emacs |

### LANGUAGE SKILLS

|          |                                                                        |
|----------|------------------------------------------------------------------------|
| FRANÇAIS | Mother tongue                                                          |
| ENGLISH  | Fluent, working operational                                            |
| MANDARIN | Good working knowledge<br>Used in daily life and working environnement |
| GERMAN   | Good working knowledge                                                 |

### EXTRA INFORMATION

- I am reviewing articles for *IEEE Transactions on Signal Processing*, *Signal Processing*, and *SIAM Journal on Mathematics of Data Science*.
- I was a visitor at Gdansk University of Technology under the supervision of Anna Jezierska in January 2020.

## PUBLICATIONS

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### Journal Articles

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1. Arthur Marmin and Frédéric J.P. Richard. **VARPROX: a primal-dual variable projection method for the minimization of penalized separable non-linear least squares.** *Inverse Probl.*, October 2025. ISSN 1361-6420. doi: 10.1088/1361-6420/ae0e48.
2. Philipp Krah, Arthur Marmin, Beata Zorawski, Julius Reiss, and Kai Schneider. **A robust shifted proper orthogonal decomposition: Proximal methods for decomposing flows with multiple transports.** *SIAM J. Sci. Comput.*, 47(2):A633–A656, March 2025. ISSN 1095-7197. doi: 10.1137/24m164392x.
3. Arthur Marmin, José Henrique de Morais Goulart, and Cédric Févotte. **Majorization-minimization for sparse nonnegative matrix factorization with the  $\beta$ -divergence.** *IEEE Trans. Signal Process.*, 71:1435–1447, April 2023. doi: 10.1109/tsp.2023.3266939.
4. Arthur Marmin, José Henrique de Morais Goulart, and Cédric Févotte. **Joint majorization-minimization for nonnegative matrix factorization with the  $\beta$ -divergence.** *IEEE Trans. Signal Process.*, 209:109048, August 2023. doi: 10.1016/j.sigpro.2023.109048.
5. Arthur Marmin, Marc Castella, Jean-Christophe Pesquet, and Laurent Duval. **Sparse signal reconstruction for nonlinear models via piecewise rational optimization.** *Signal Process.*, 179:107835, February 2021. doi: 10.1016/j.sigpro.2020.107835.
6. Arthur Marmin, Anna Jezierska, Marc Castella, and Jean-Christophe Pesquet. **Global Optimization for Recovery of Clipped Signals Corrupted With Poisson-Gaussian Noise.** *IEEE Signal Process. Lett.*, 27:970–974, May 2020. doi: 10.1109/lsp.2020.2998699.
7. Marc Castella, Jean-Christophe Pesquet, and Arthur Marmin. **Rational Optimization for Nonlinear Reconstruction with Approximate  $\ell_0$  Penalization.** *IEEE Trans. Signal Process.*, 67(6):1407–1417, March 2019. doi: 10.1109/tsp.2018.2890065.

### Conference Articles

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6. Beata Zorawski, Shubhaditya Burela, Philipp Krah, Arthur Marmin, and Kai Schneider. **Automated transport separation using the neural shifted proper orthogonal decomposition.** In *The 9th European Congress on Computational Methods in Applied Sciences and Engineering*. CIMNE, 2024. doi: 10.23967/eccomas.2024.066.
7. Elie Leroy, Arthur Marmin, Marc Castella, and Laurent Duval. **Weight Identification Through Global Optimization in a New Hysteretic Neural Network Model.** In *Proc. Int. Conf. Acoust. Speech Signal Process.* IEEE, June 2021. doi: 10.1109/icassp39728.2021.9413383.
8. Arthur Marmin, Marc Castella, and Jean-Christophe Pesquet. **Globally Optimizing Owing to Tensor Decomposition.** In *Proc. European Signal Processing Conference*, pages 990–994. IEEE, January 2021. doi: 10.23919/eusipco47968.2020.9287511.
9. Arthur Marmin, Marc Castella, and Jean-Christophe Pesquet. **Robust Reconstruction with Nonconvex Subset Constraints: A Polynomial Optimization Approach.** In *IEEE Int. Workshop Mach. Learn. Signal Process.* IEEE, September 2020. doi: 10.1109/mlsp49062.2020.9231524.
10. Arthur Marmin, Marc Castella, and Jean-Christophe Pesquet. **A Moment-Based Approach for Guaranteed Tensor Decomposition.** In *Proc. Int. Conf. Acoust. Speech Signal Process.*, pages 3927–3931. IEEE, May 2020. doi: 10.1109/icassp40776.2020.9054186.
11. Arthur Marmin, Marc Castella, and Jean-Christophe Pesquet. **Detecting the Rank of a Symmetric Tensor.** In *Proc. European Signal Processing Conference*, pages 1–5. IEEE, September 2019. doi: 10.23919/eusipco.2019.8902781.
12. Arthur Marmin, Marc Castella, and Jean-Christophe Pesquet. **Sparse Signal Reconstruction with a Sign Oracle.** In *Proc. Signal Processing with Adaptive Sparse Structured Representations (SPARS) workshop*, July 2019.
13. Arthur Marmin, Marc Castella, and Jean-Christophe Pesquet. **How to Globally Solve Non-convex Optimization Problems Involving an Approximate  $\ell_0$  Penalization.** In *Proc. Int. Conf. Acoust. Speech Signal Process.*, pages 5601–5605. IEEE, May 2019. doi: 10.1109/icassp.2019.8683692.
14. Arthur Marmin, Marc Castella, Jean-Christophe Pesquet, and Laurent Duval. **Signal Reconstruction from Sub-sampled and Non-linearly Distorted Observations.** In *Proc. European Signal Processing Conference*, pages 1970–1974. IEEE, September 2018. doi: 10.23919/eusipco.2018.8553174.
15. Arthur Marmin, Chun-Hung Lai, Haruyuki Tago, Hsun-Lun Huang, and Juin-Ming Lu. **Architecture agnostic energy model for GPU-based design.** In *2016 International Symposium on VLSI Design, Automation and Test (VLSI-DAT)*. IEEE, April 2016. doi: 10.1109/vlsi-dat.2016.7482557.